



Applied Data Mining

**— Week 1—
Spring 2026**

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Course Detail

Text Books:

- Jiawei Han & Micheline Kamber, Jian Pei (2011). Data Mining: Concepts and Techniques, 3rd Edition.
- Pang-Ning Tan, Michael Steinbach, and Vipin Kumar (2005). Introduction to Data Mining.
- Charu C. Aggarwal (2015). Data Mining: The Textbook
- D. Hand, H. Mannila, P. Smyth (2001). Principles of Data Mining. MIT Press.

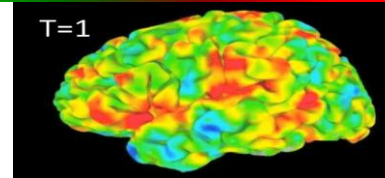
Why Data Mining? Commercial Viewpoint

- Lots of data is being collected and warehoused
 - Web data
 - Google has Peta Bytes of web data
 - Facebook has billions of active users
 - purchases at department/grocery stores, e-commerce
 - Amazon handles millions of visits/day
 - Bank/Credit Card transactions
- Computers have become cheaper and more powerful
- Competitive Pressure is Strong
 - Provide better, customized services for an edge (e.g. in Customer Relationship Management)



Why Data Mining? Scientific Viewpoint

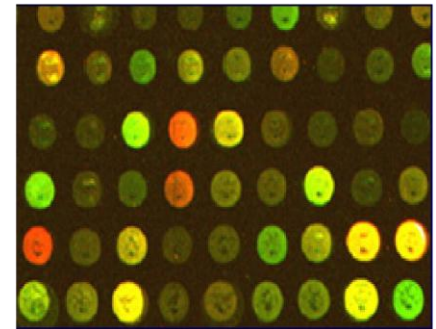
- Data collected and stored at enormous speeds
 - Remote sensors on a satellite
 - NASA Earth Observing System Data and Information System (EOSDIS) archives over petabytes of earth science data / year
 - Telescopes scanning the skies
 - Sky survey data
 - High-throughput biological data
 - Scientific simulations
 - Terabytes of data generated in a few hours
- Data mining helps scientists
 - In automated analysis of massive datasets
 - In hypothesis formation



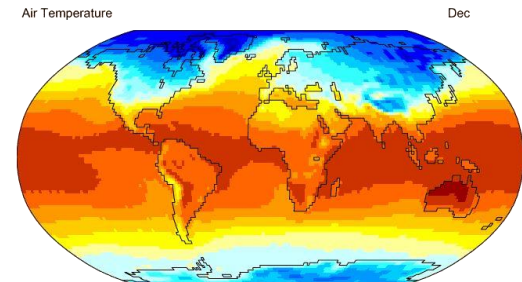
fMRI Data from Brain



Sky Survey Data



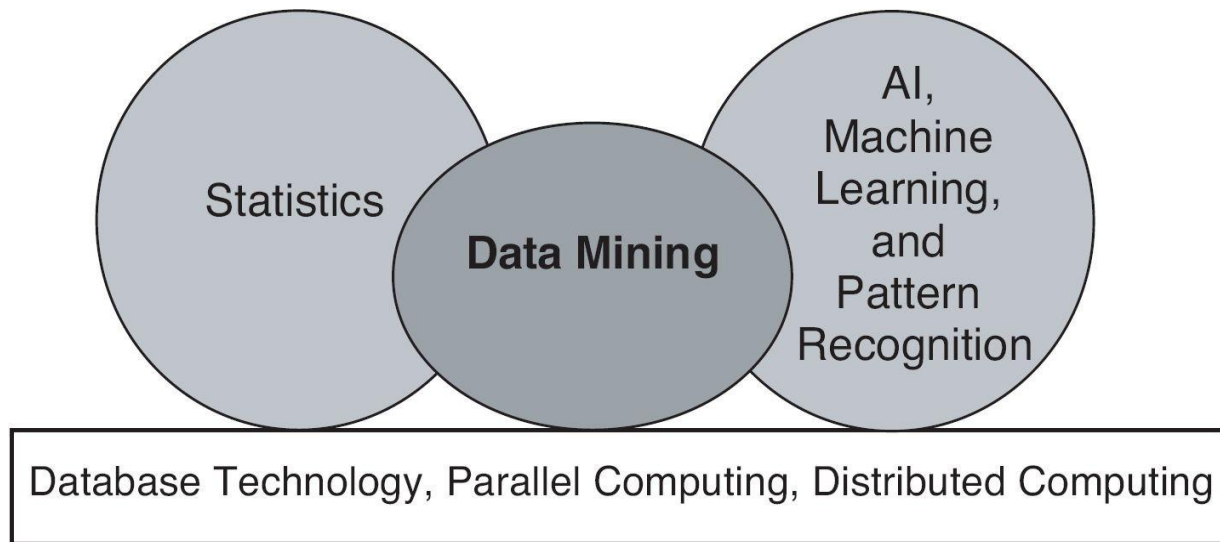
Gene Expression Data



Surface Temperature of Earth

Origins of Data Mining

- Draws ideas from machine learning/AI, pattern recognition, statistics, and database systems
- Traditional techniques may be unsuitable due to data that is
 - Large-scale
 - High dimensional
 - Heterogeneous
 - Complex
 - Distributed



- A key component of the emerging field of data science and data-driven discovery

What is Data Mining?

- Many Definitions

- Non-trivial extraction of implicit, previously unknown and potentially useful information from data
- Exploration & analysis, by automatic or semi-automatic means, of large quantities of data in order to discover meaningful patterns
- Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data (Knowledge discovery from data)

What is Data Mining?

- Alternative names
 - Knowledge discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging (searching), information harvesting, business intelligence, etc.
- Watch out: Is everything “data mining”?
 - Simple search and query processing

What is Data Science

- ❑ **Data science** is lying at the intersection of computer science, statistics, and substantive application domains.
- ❑ It is the study of “Where information comes from, what it represents and how it can be turned into a valuable resource in the creation of business and IT strategies”
- ❑ **Data science** is a "concept to unify/merge statistics, data analysis, machine learning and their related methods" in order to "understand and analyze actual phenomena" with data”.

What is Machine Learning

- The field of machine learning is concerned with the question of how to construct computer programs that automatically improve with experience (T. Mitchell).

OR

- Principles, methods, and algorithms for learning and prediction on the basis of past experience.

OR

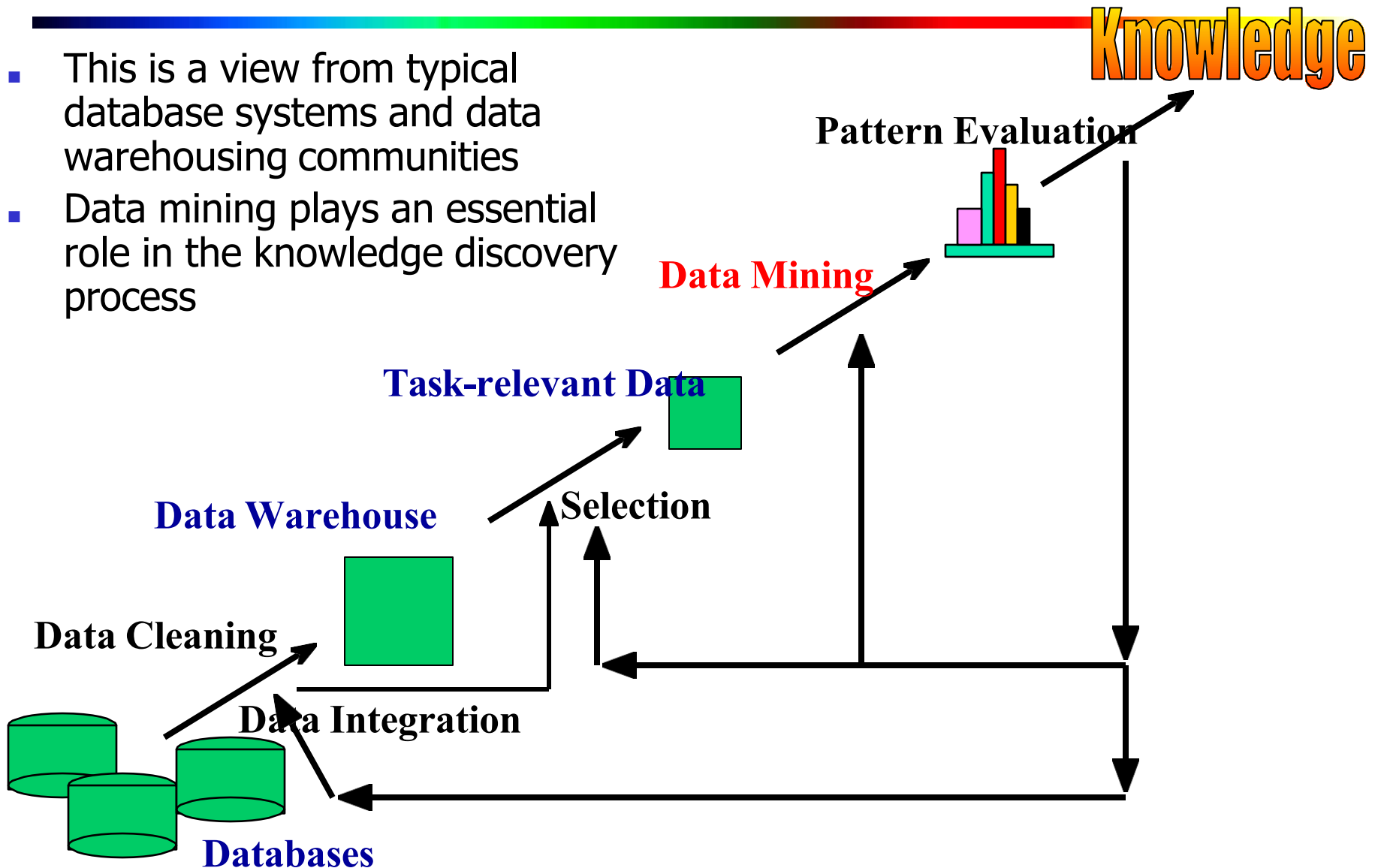
- In the broadest sense, any method that incorporates information from training samples in the design of a classifier employs learning.

Difference b/w Data Mining and Data Science

Data Mining	Data Science
Data mining is a process of extracting useful information, patterns, and trends from huge databases.	Data science refers to the process of obtaining valuable insights from structured and unstructured data by using various tools and methods.
Data mining is a technique.	Data science is a field.
Primarily used for business purposes.	Primarily used for scientific purposes.
It is involved with the process.	It emphasizes the science of the data.
Data mining aims to make data more important and usable; it means extracting only useful information.	The objective of data science is to create a dominant data product.
Data mining is a technique that is a part of KDD (Knowledge discovery in database process).	It is related to the field of study like Mechanical engineering, Cloud architecture, etc.
It primarily deals with structured data.	It deals with any kind of data like structured, semi-structured, and unstructured.

Knowledge Discovery (KDD) Process

- This is a view from typical database systems and data warehousing communities
- Data mining plays an essential role in the knowledge discovery process



Knowledge Discovery (KDD) Process

- **Data cleaning:**

To remove noise and inconsistent data

- **Data integration:**

Where multiple data sources may be combined

- **Data selection:**

Where data relevant to the analysis task are retrieved from the database

- **Data transformation:**

Where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations

Knowledge Discovery (KDD) Process

- **Data mining:**

An essential process where intelligent methods are applied to extract data patterns

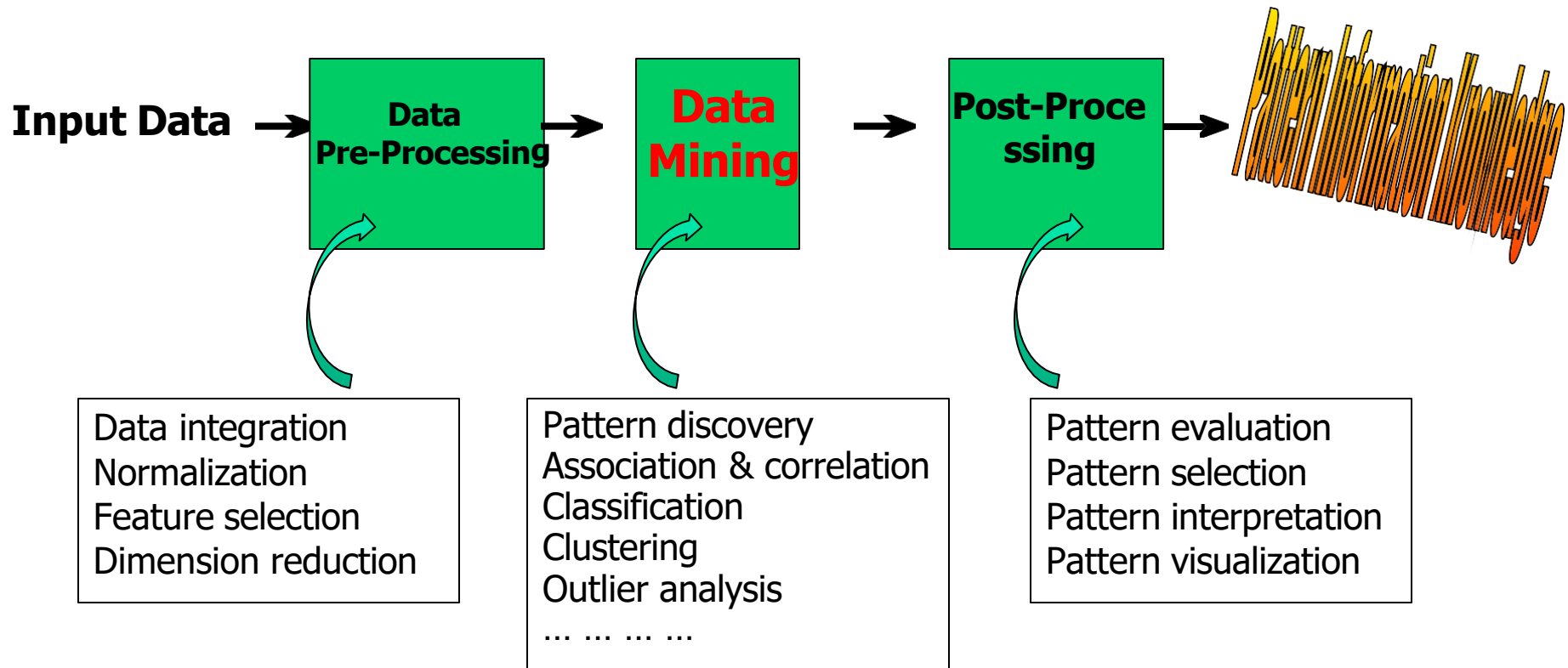
- **Pattern evaluation:**

To identify the truly interesting patterns representing knowledge based on interestingness measures

- **Knowledge presentation :**

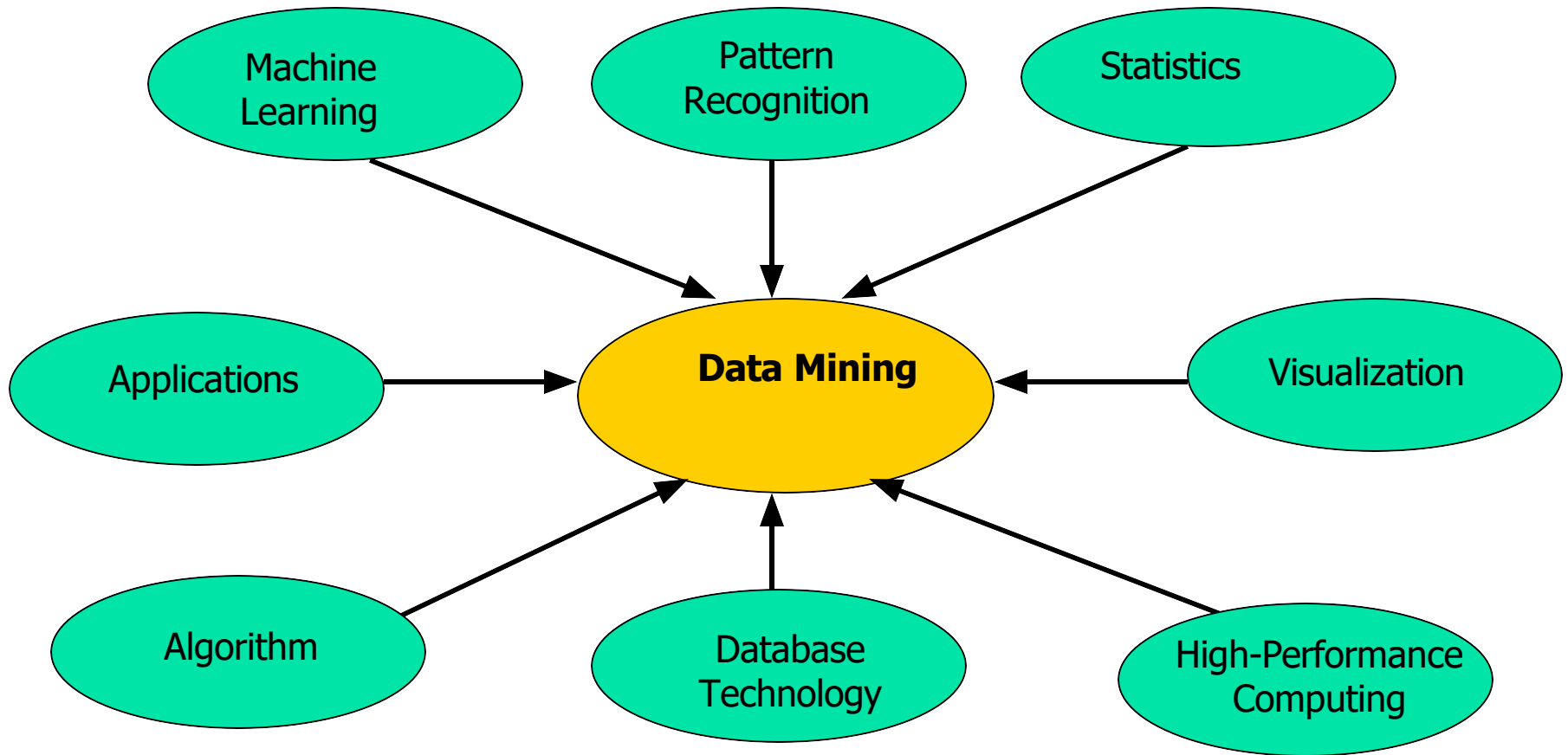
Where visualization and knowledge representation techniques are used to present mined knowledge to users

KDD Process: A Typical View from ML and Statistics



- This is a view from typical machine learning and statistics communities

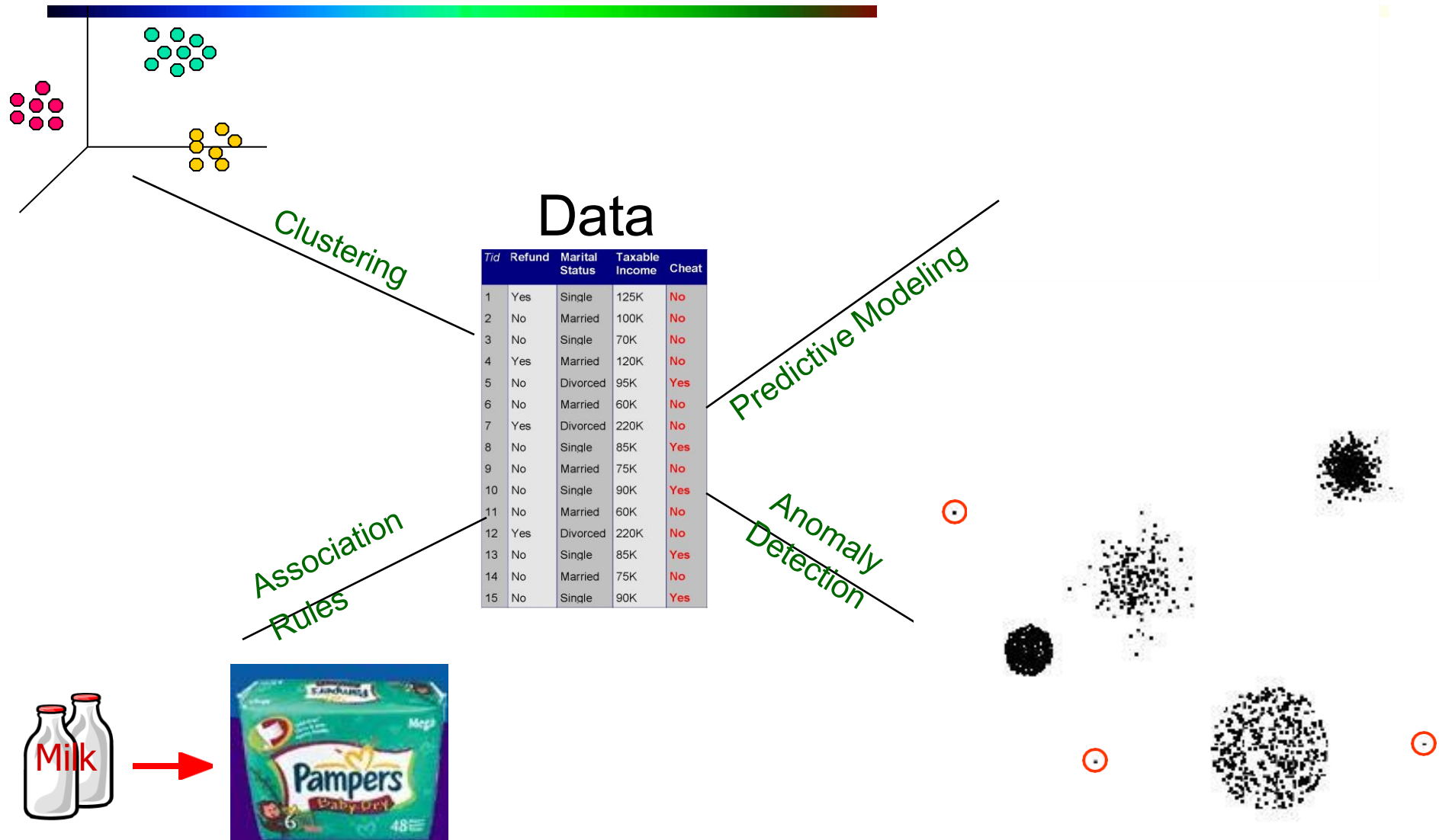
Data Mining: Confluence of Multiple Disciplines



Data Mining Tasks

- Prediction Methods
 - Use some variables to predict unknown or future values of other variables.
- Description Methods
 - Find human-interpretable patterns that describe the data.

Data Mining Tasks ...

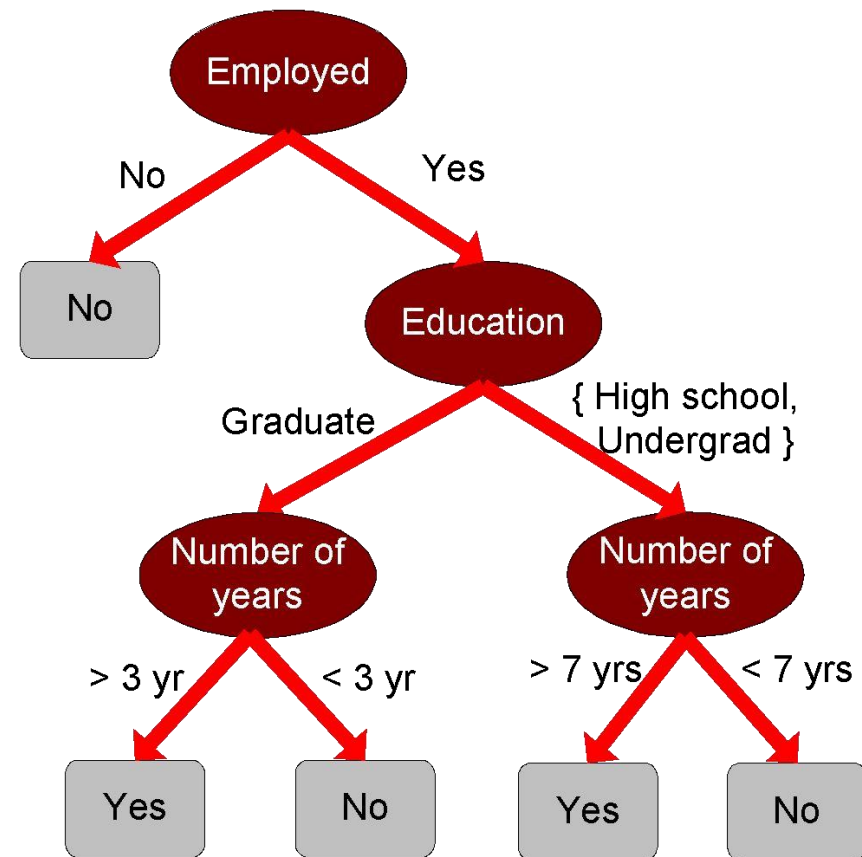


Predictive Modeling: Classification

- Find a model for class attribute as a function of the values of other attributes
- Model for predicting credit worthiness

Class

<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
...	...	...	...	...

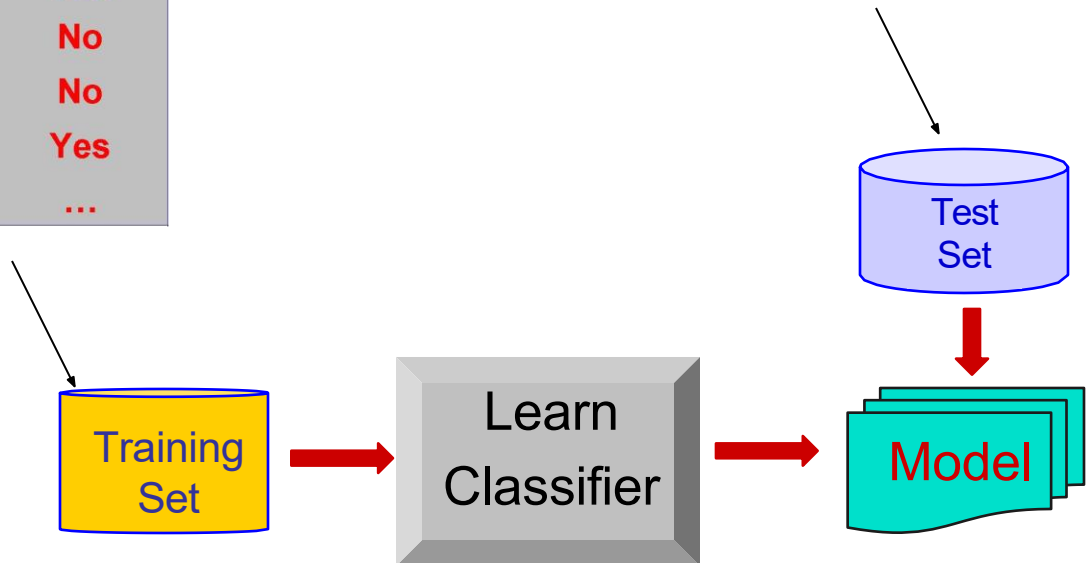


Classification Example

categorical categorical quantitative class

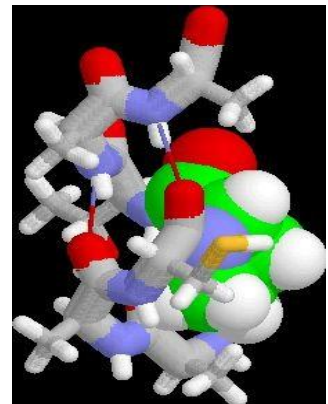
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4	Yes	High School	10	Yes
...	...	...	...	...

<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Undergrad	7	?
2	No	Graduate	3	?
3	Yes	High School	2	?
...	...	...	...	...



Examples of Classification Task

- ❑ Classifying credit card transactions as legitimate or fraudulent
- ❑ Classifying land covers (water bodies, urban areas, forests, etc.) using satellite data
- ❑ Categorizing news stories as finance, weather, entertainment, sports, etc
- ❑ Identifying intruders in the cyberspace
- ❑ Predicting tumor cells as benign or malignant
- ❑ Classifying secondary structures of protein as alpha-helix, beta-sheet, or random coil



Classification: Application 1

- Fraud Detection
 - **Goal:** Predict fraudulent cases in credit card transactions.
 - **Approach:**
 - Use credit card transactions and the information on its account-holder as attributes.
 - When does a customer buy, what does he buy, how often he pays on time, etc
 - Label past transactions as fraud or fair transactions. This forms the class attribute.
 - Learn a model for the class of the transactions.
 - Use this model to detect fraud by observing credit card transactions on an account.

Regression

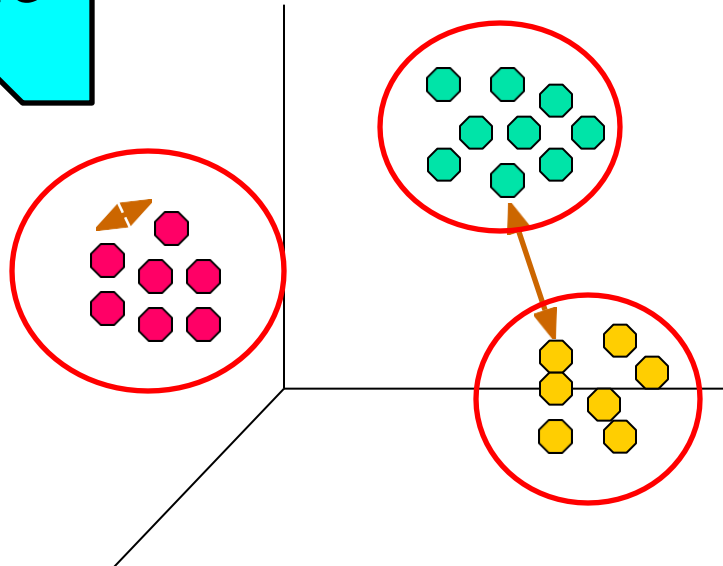
- Predict a value of a given continuous valued variable based on the values of other variables, assuming a linear or nonlinear model of dependency.
- Extensively studied in statistics, neural network fields.
- Examples:
 - Predicting sales amounts of new product based on advertising expenditure.
 - Predicting wind velocities as a function of temperature, humidity, air pressure, etc.
 - Time series prediction of stock market indices.

Clustering

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups

Intra-cluster
distances are
minimized

Inter-cluster
distances are
maximized



Applications of Cluster Analysis

■ Understanding

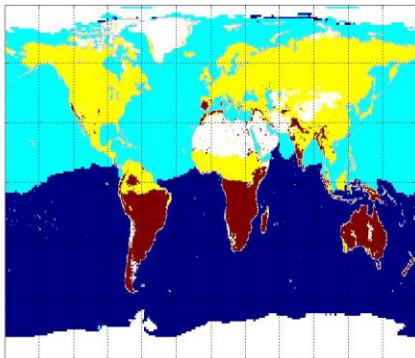
- Custom profiling for targeted marketing
- Group related documents for browsing
- Group genes and proteins that have similar functionality
- Group stocks with similar price fluctuations

■ Summarization

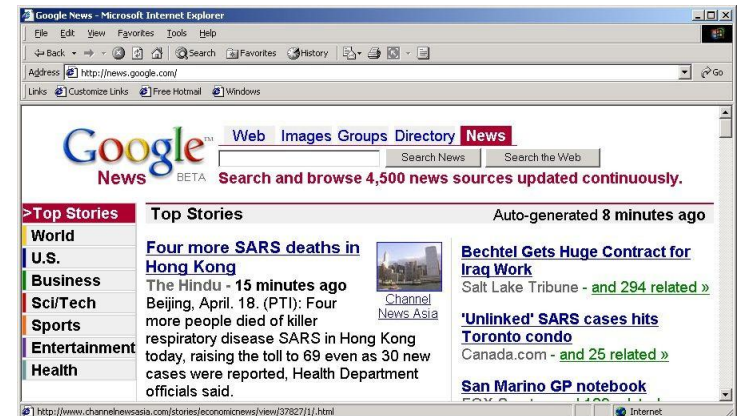
- Reduce the size of large data sets



Courtesy: Michael Eisen



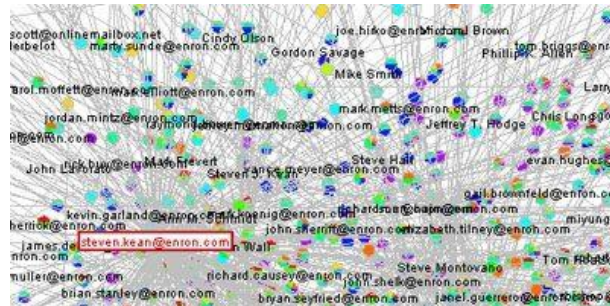
Use of K-means to partition Sea Surface Temperature (SST) and Net Primary Production (NPP) into clusters that reflect the Northern and Southern Hemispheres.



Clustering: Application 1

- Document Clustering:
 - **Goal:** To find groups of documents that are similar to each other based on the important terms appearing in them.
 - **Approach:** To identify frequently occurring terms in each document. Form a similarity measure based on the frequencies of different terms. Use it to cluster.

Enron email dataset



Association Rule Discovery: Definition

- Given a set of records each of which contain some number of items from a given collection
 - Produce dependency rules which will predict occurrence of an item based on occurrences of other items.

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

$\{\text{Milk}\} \rightarrow \{\text{Coke}\}$

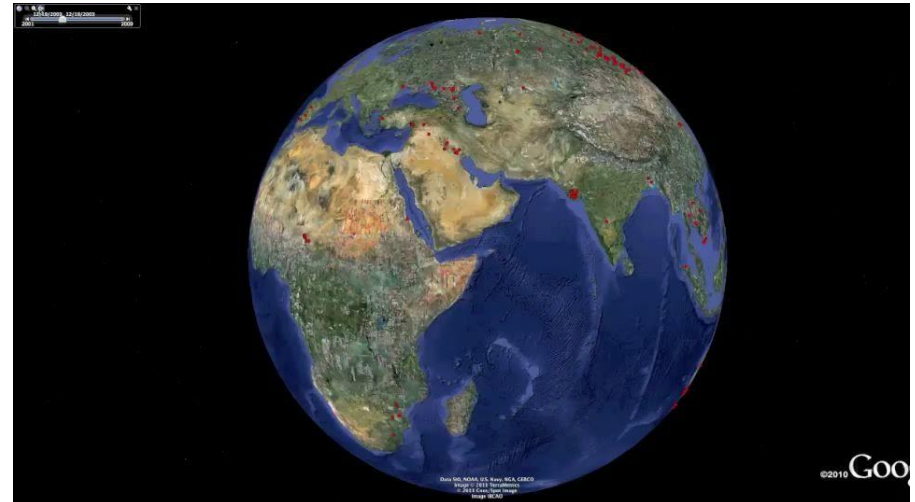
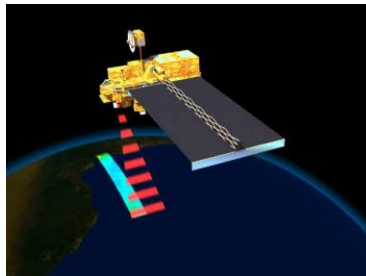
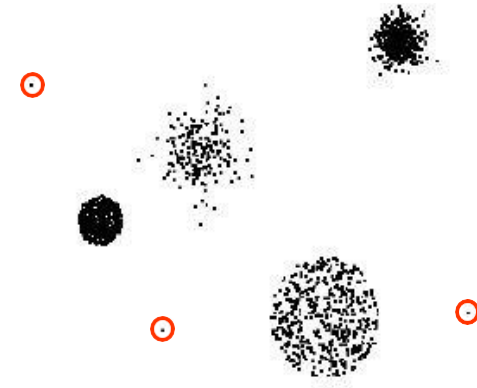
$\{\text{Diaper, Milk}\} \rightarrow \{\text{Beer}\}$

Association Analysis: Applications

- Market-basket analysis
 - Rules are used for sales promotion, shelf management, and inventory management
- Telecommunication alarm diagnosis
 - Rules are used to find combination of alarms that occur together frequently in the same time period
- Medical Informatics
 - Rules are used to find combination of patient symptoms and test results associated with certain diseases

Deviation/Anomaly/Change Detection

- Detect significant deviations from normal behavior
- Applications:
 - Credit Card Fraud Detection
 - Network Intrusion Detection
 - Identify anomalous behavior from sensor networks for monitoring and surveillance.
 - Detecting changes in the global forest cover.



Data Mining: On What Kinds of Data?

- Database-oriented data sets and applications
 - Relational database, data warehouse, transactional database
- Advanced data sets and advanced applications
- **Data streams** (e.g., video surveillance and sensor data, which are continuously transmitted).
- **Time-series data, temporal data, sequence data** (e.g., historical records, stock exchange data, and time-series and biological sequence data).
- **Engineering design data** (e.g., the design of buildings, system components, or integrated circuits).

Data Mining: On What Kinds of Data?

- Advanced data sets and advanced applications
 - **Structure data, graphs, social networks and multi-linked data**
 - **Object-relational databases**
 - **Spatial data and spatiotemporal data** (e.g. maps)
 - **Hypertext and Multimedia** (including text, image, video, and audio data)
- **The World-Wide Web** (a huge, widely distributed information repository made available by the Internet).



Any Question